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Full Length Article

# Learned nonlocal feature matching and filtering for RAW image denoising

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ABSTRACT

Most modern image denoising networks rely on large depth and parameter counts, making limited use of technical knowledge brought by classical model-based denoisers. This poses a challenge not only for understanding their properties, but also for deploying them on real devices which may present resource constraints and diverse noise profiles. Tackling both issues, we propose a RAW-to-RAW denoising architecture that incorporates the structure of classical self-similarity-based denoisers into a fully learnable neural network. Our design centers on a novel nonlocal block that parallels the established pipeline of neighbor matching, collaborative filtering and aggregation popularized by nonlocal patch-based methods, operating on learned multiscale feature representations. This built-in nonlocality efficiently expands the receptive field, so that a single block per scale with a moderate number of neighbors is sufficient to obtain high-quality results. Training the network on a curated dataset with clean real RAW data and modeled synthetic noise while conditioning it on a noise level map yields a sensor-agnostic denoiser that generalizes effectively to unseen devices. Both quantitative and visual results on benchmarks and in-the-wild photographs position our method as a practical solution for real-world RAW denoising, achieving results competitive with state-of-the-art convolutional and transformer-based denoisers while having significantly fewer parameters and a faster runtime. The code is available at https://github.com/MIA-UIB/nonlocal-matchfilter.

## 1. Introduction

The presence of noise in digital images is inevitable due to the very own workings of the image formation process. Camera sensors, as well as more complex image acquisition systems, are susceptible to undesirable measurement errors that arise from multiple sources. These include the variability caused by the random nature of photon counting, thermal fluctuations in the electronic circuits of the sensors, and conversion to digital signals (Wei et al., 2022), among other setting-specific perturbations. While modern advances in sensor technology have allowed for a reduction of noise magnitude at the physical level (Boukhayma et al., 2016), it still remains a fundamental problem that must be dealt with after image capture.

Noise obscures fine details and textures, deteriorating signal information. In a world where visual quality is paramount for phone camera users, its presence in day-to-day pictures is not acceptable, except for specific artistic purposes. It also hinders interpretation in diagnostic-oriented images (e.g. medical images), and compromises the performance of downstream computer vision tasks (Pei et al., 2021). For all these reasons, effective denoising methods that faithfully recover the underlying scene structure become essential for ensuring that images meet quality standards expected in modern imaging applications.

Over the years, a significant share of work has been dedicated to studying the removal of Additive White Gaussian Noise (AWGN) (Elad et al., 2023), whose simplicity makes it an ideal testbed for denoising methods. In modern digital photography, however, denoising must address noise characteristics that differ substantially from the AWGN model. At the sensor level, the combination of the two primary sources of noise (shot and readout noise) follows a compound Poisson-Gaussian distribution. Its approximation by a heteroskedastic Gaussian with signal-dependent variance is often used as the simplest noise model for RAW images (Foi et al., 2008; Healey & Kondepudy, 1994), with more complex approaches proposed for low-light scenarios (Cao et al., 2023; Feng et al., 2024; Lu et al., 2025; Wei et al., 2022; Zhang et al., 2023, 2021). Critically, noise parameters in such models can vary significantly across devices and camera settings. This motivates the use of noise-aware RAW denoisers that use prior noise level information to adapt to sensor characteristics.

Denoising algorithms can be broadly categorized into model-based and data-driven approaches. Classical model-based methods rely on

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meticulously hand-crafted priors that capture natural image statistics. Among these, patch-based methods exploiting nonlocal self-similarity have had the most influence in the field (Buades et al., 2005). Perhaps the most widely known is the seminal BM3D algorithm (Dabov et al., 2007), which spearheaded a family of denoisers with a common three-step pipeline: matching similar patches, transforming them into a suitable domain for filtering, and aggregating them back. While these methods usually assume Gaussian noise, using its variance to adjust the strength of the filtering, they can be extended to the sensor-specific distributions of the RAW domain by using variance stabilization transforms (Mäkitalo & Foi, 2014) or local variance estimators (Sánchez-Beeckman et al., 2026). Although they have been overshadowed by neural networks, classical methods remain relevant today, as their strong theoretical foundations and explicit structure provide insight into how different image features are processed.

In contrast, data-driven approaches based on deep learning, beginning with convolutional neural networks (CNNs) like DnCNN (Zhang et al., 2017a) and later with transformers (Liang et al., 2021), have shown excellent performance by learning to denoise directly from large datasets, up to the point of surpassing classical methods in quality. By accepting a noise level map as input, networks like FFDNet (Zhang et al., 2018) and DRUNet (Zhang et al., 2022) allow some degree of control over the filtering strength, an idea that has also been applied with success in the RAW domain (Li et al., 2024b). However, even these noise-aware networks remain largely opaque, as their deep end-to-end learned representations obscure how they use image structures to remove noise.

This opacity is partly rooted in the architectural choices that uphold most modern denoisers, which use little domain knowledge cultivated before the ubiquity of deep learning. Convolutional networks, whose layers operate locally by design, cannot exploit the nonlocal self-similarity that proved so effective in classical methods. Transformers, via their attention mechanism, do perform data-dependent weighted averaging that evokes classical nonlocal adaptive filters (Milanfar, 2013), yet their use entails many pixel comparisons—a significant portion of which may contribute only marginally to the result (Cherel et al., 2024)—and stacking many layers to attain high-quality results. This reliance on depth and scale limits interpretability and can also hinder deployment on devices with computation and memory constraints, where RAW denoising often takes place. Although some efforts have been made to incorporate more classical notions of nonlocal self-similarity into denoising networks (Cruz et al., 2018; Lefkimmiatis, 2017; Meng et al., 2024; Yan et al., 2020), most of them embed classical priors within otherwise generic architectures rather than building new architectures around these notions.

In this work, we propose a deep neural network for RAW-to-RAW image denoising that explicitly translates the interpretable structure of classical nonlocal methods (in the sense that one may understand each component's function and effect) into an end-to-end learnable framework. Our architecture is built around a novel Nonlocal Feature Matching and Filtering block that mirrors the three-stage pipeline of classical collaborative filtering methods. Crucially, we operate on learned feature representations within a multiscale architecture rather than directly on image patches, allowing the network to leverage richer semantic information while maintaining the core principle of nonlocal self-similarity. By also accepting a noise level map as input, the network adapts to spatially-varying noise, making it naturally suited for realistic camera noise models. The network learns to identify matching neighboring features within a search window around each pixel for collaborative filtering. This built-in nonlocality, combined with the multiscale structure, expands the receptive field efficiently without requiring the computational overhead of self-attention or excessive network depth. The result is a denoising network that bridges classical methods with modern deep learning, retaining the theoretical grounding from the former while harnessing the representational power of the latter.

Our main contributions are as follows:

- We propose a novel, fully learnable neural module that performs nonlocal block matching, collaborative filtering and aggregation in an adapted feature domain, providing an alternative to black-box image denoising architectures.
- We model the noise profiles of a variety of camera sensors, using them to build a dataset with clean real RAW data and on-demand synthetic noise with which to train a network for sensor-agnostic RAW-to-RAW image denoising.
- We demonstrate that integrating the proposed block within a three-scale UNet achieves high-quality results that are comparable or superior to modern methods with deeply-stacked generic layers, all while using significantly fewer parameters and less inference time.
- We test our network on unseen sensors: the DND RAW benchmark (Plötz & Roth, 2017), achieving state-of-the-art quantitative results, and in-the-wild phone captures, also with strong qualitative performance.

## 2. Related work

### 2.1. Classical self-similarity-based denoising

Image restoration has historically relied on mathematical models of the intrinsic characteristics of images and the relations that arise between their pixels. Among classical methods, patch-based ones exploiting the self-similarity of images to denoise them have had the most long-lasting repercussion. Nonlocal Means (Buades et al., 2005) prompted this idea by proposing to eliminate noise by matching and averaging patches without explicitly encouraging their spatial proximity. Kervrann and Boulanger (2006) introduced spatial adaptation to it by refining patch weights based on a local estimation of their noise variance. Deviating from kernel-based averaging, BM3D (Dabov et al., 2007) became the reference patch-based algorithm by using collaborative Wiener filtering in a fixed orthonormal basis. The method popularized the general three-step architecture of patch matching, filtering and aggregation, which was subsequently used by algorithms like PLOW (Chatterjee & Milanfar, 2012), NL-Bayes (Lebrun et al., 2013), and WNNM (Gu et al., 2014), each employing different collaborative filtering strategies to exploit priors such as joint sparsity and low-rank signal representations.

Despite their initial success, classical methods are limited by their reliance on hand-crafted similarity metrics and filters, which may not capture the full complexity of natural images. Nonetheless, the stated three-step structure remains a powerful paradigm for denoising, which this work integrates into a more versatile learnable counterpart.

### 2.2. Denoising CNNs

The arrival of CNNs caused a paradigm shift in the field of denoising. DnCNN (Zhang et al., 2017a) showed that a simple feed-forward network composed of sequential convolutional layers, ReLU activations and batch normalization (Ioffe & Szegedy, 2015) could outperform classical methods considerably when trained end-to-end to learn the residual between clean and noisy images, despite working only locally. The same year, IRCNN (Zhang et al., 2017b) used dilated convolutions to enlarge the receptive field, although at the cost of generating artifacts around sharp edges. FFDNet (Zhang et al., 2018) enlarged it in an alternative way: it applied DnCNN to four downsampled subimages obtained via pixel unshuffle (Shi et al., 2016), also concatenating a noise map to the input so as to help it generalize to multiple noise levels.

Multiscale processing has since become widely adopted to aggregate distant information and alleviate the limitations imposed by local receptive fields. SADNet (Chang et al., 2020) uses a spatial-adaptive block based on deformable convolutions (Dai et al., 2017) within a multiscale encoder-decoder architecture. Through a neural architecture search, CLEARER (Gou et al., 2020) learns when and how to extract and fuse cross-scale features. Taking a different, interpretable approach, DeamNet (Ren et al., 2021) unfolds a model-based denoiser

**Table 1**
Comparison between our proposed method and the closest nonlocal denoising neural networks in the literature.

| Method | Matching | Filtering | Aggregation | Network integration |
|---|---|---|---|---|
| Lefkimmiatis (2017) | Window search<br>Image patches | Depthwise conv &<br>grouped channel conv | Neighbor indexing | Variational unrolling |
| Qiao et al. (2017) | Classical KNN<br>Image patches | Learned sparse matrices & radial basis functions | | Variational unrolling |
| Plötz and Roth (2018) | Differentiable KNN<br>Learned features | DnCNN | Neighbor indexing | Feed-forward network |
| Liu et al. (2018) | Correlation weights<br>Learned features | Per-pixel window self-attention | | Recurrent network |
| Meng et al. (2024) | Correlation weights<br>Learned features | Shifted window self-attention | | Variational unrolling |
| Ours | Offset CNN<br>Learned features | Grouped channel conv &<br>modulation map | 1x1 conv | 3-scale UNet |

with an adaptive consistency prior into a multiscale end-to-end trainable convolutional network. MSANet (Gou et al., 2022) uses an asymmetric encoder-decoder architecture with different subnetworks per scale, fusing coarse details into finer scales with modulated deformable convolutions (Zhu et al., 2019). Also taking a noise map as input, DRUNet (Zhang et al., 2022) achieves state-of-the-art results with a bias-free four-scale UNet (Ronneberger et al., 2015) purely composed of residual blocks (He et al., 2016), which the authors use in a Plug-and-Play scheme for different restoration tasks.

Even so, locality is still a fundamental limitation of purely convolutional architectures, as they cannot exploit image self-similarity the way classical methods do. Our proposal addresses this shortcoming, incorporating a nonlocal mechanism into an otherwise simple CNN-based UNet architecture with a noise map.

### 2.3. Learning from nonlocal information

Noting the resemblance between the nonlocal means filter and the self-attention mechanism used in language models (Vaswani et al., 2017), Wang et al. (2018) proposed a general feed-forward block that weights features based on their relationships across distant locations. This mechanism became the keystone in the development of nonlocal attention networks (Mei et al., 2023; Zhang et al., 2019) and vision transformers (Dosovitskiy et al., 2021) as alternatives to CNNs. Since then, transformers have become state-of-the-art in image restoration (Chen et al., 2021; Li et al., 2024a; Yin & Ma, 2022; Zhou et al., 2024; Zhuge et al., 2023).

Despite their strong performance, vision transformers are computationally expensive. Starting with SwinIR (Liang et al., 2021), architectures based on shifted windowing schemes have appeared so as to reduce their cost, restricting self-attention to nonoverlapping local regions that are shifted after each layer (Tian et al., 2024a; Wang et al., 2022). As an alternative, Restormer (Zamir et al., 2022) proposes a global transposed attention transformer block with a depthwise convolutional head, which is also embedded in a UNet architecture. Ensuing state-of-the-art denoisers have since largely been based on these two foundational configurations, with variations mainly on auxiliary modules, depth, mixing of convolutional and transformer blocks, and training strategies (Hu et al., 2026; Huang & Huang, 2025; Tian et al., 2024b; Zhang et al., 2024).

Apart from transformers, some parallel works have explored designing networks with other explicit nonlocal mechanisms. Liu et al. (2018) were the first to build a recurrent network with nonlocal operations, using soft matching on feature representations. Lefkimmiatis (2017) proposed applying block matching on noisy image patches and feeding them to a CNN. Qiao et al. (2017) also used block matching to embed a nonlocal self-similarity prior into the TNRD network (Chen & Pock, 2017). Plötz and Roth (2018) designed a learnable block based on a differentiable relaxation of $K$-nearest neighbors selection, interleaving it with DnCNN modules. More recently, Meng et al. (2024) extended this direction by combining correlation-based matching within shifted windows with a variational unrolling framework.

Still, transformers have become the preferred option for high quality denoising, as evidenced by their dominance in the latest benchmarks (Sun et al., 2025). This has left other nonlocal paradigms underexplored in recent literature. In this work, by returning to explicit matching and building a network around a classical collaborative filtering pipeline, we show that such approaches remain viable, competitive, and more time-efficient than state-of-the-art transformers. We summarize the key architectural differences between our approach and prior block-matching-based methods in Table 1.

### 2.4. Denoising RAW images

Not only the sensor noise model, but the distinct characteristics of RAW image data demand dedicated denoising strategies. Unlike typical RGB images, RAW data has a mosaic structure—caused by a Color Filter Array (CFA)—in which spatial and chromatic information are interleaved. Classical methods have been adapted to these CFA patterns by packing same-color pixels into separate channels and using color decorrelation transforms (Akiyama et al., 2015; Buades & Duran, 2020; Zhang et al., 2009), or by jointly denoising and demosaicking (Chatterjee et al., 2011; Hirakawa & Parks, 2006; Tan et al., 2017). While Gaussian denoisers can be applied after variance stabilization to remove sensor noise (Mäkitalo & Foi, 2014), the transformed noise distribution often exhibits heavier tails than a true Gaussian (Zhang et al., 2015), resulting in inaccurate restoration under low-light conditions.

The emergence of deep learning has transformed the problem from one of algorithmic adaptation to one of data acquisition. Several datasets have been introduced to meet this need. To deal with especially problematic low signal-to-noise ratios in darker regions, works from Chen et al. (2018), Prabhakar et al. (2021) and Wei et al. (2022) present datasets composed of low-light scenes accompanied with long-exposure clean images. Brummer and Vleeschouwer (2025) share a diverse collection of paired RAW images from a variety of camera models and CFA patterns. For benchmarking, DND (Plötz & Roth, 2017) and SIDD (Abdelhamed et al., 2018) provide noisy images and a platform to evaluate denoised results against their held-out ground truths. RAW denoising datasets have also been curated for the case of video, as is the case of CRVD (Yue et al., 2020) and its successor ReCRVD (Yue et al., 2025).

Real data is often insufficient for effectively training RAW denoising networks, so synthesizing realistic noisy images becomes necessary. Capitalizing on the abundance of noise-free sRGB image datasets, Brooks et al. (2019) invert the image processing pipeline to do so, and train a CNN on that unprocessed data. Using a similar idea, CycleISP (Zamir et al., 2020) learns simultaneously to generate synthetic RAW data from RGB images, and to denoise and process RAW images into RGB with a two-branch convolutional network. Pseudo-ISP (Cao et al., 2024) follows the same direction, removing the need for RAW-RGB pairs to train the synthesis branch and generating pseudo-RAW images instead.

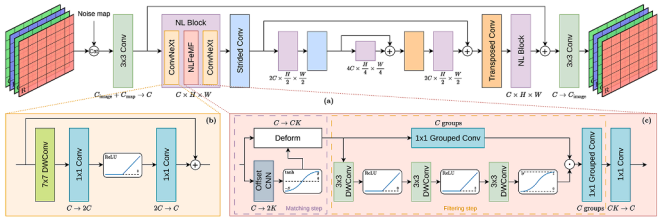


**Fig. 1.** Architecture of the proposed denoising network. We use a three-scale UNet with a single NL block per scale. The NL block envelops a Nonlocal Feature Matching and Filtering block between two simplified ConvNeXt layers.

DualDn (Li et al., 2024b) introduces a differentiable processing chain to perform dual-domain denoising in RAW and sRGB, adapting to both sensor noise and image signal processor variations. In our case, we also synthesize training data, but we limit the synthesis to the noise and add it to real clean RAW images to avoid domain shift errors caused by unprocessing.

## 3. Method

We design our network for RAW-to-RAW denoising of images captured with a Bayer CFA. To preserve their mosaic structure, we pack the RAW images into four $RG_rBG_b$ channels, and then normalize them to the $[0, 1]$ range by removing their black level offset and dividing by their saturation value. We feed the data to the network together with a noise map of the same size, concatenated along the channel dimension. The map holds the standard deviation of the noise at each pixel, estimated from the shot and readout coefficients of the sensor assuming a regular affine variance model with respect to the intensity values. Since each channel of the packed image corresponds to a distinct position, this noise map also has four channels; this yields an eight-channel input to the network, which outputs a four-channel denoised packed RAW image that can be subsequently unpacked.

### 3.1. Overall network architecture

Inspired by the structure of classical nonlocal patch-based denoisers, we propose a nonlocal neural network that learns to match self-similar image features and filters them across scales. Its overall architecture is illustrated in Fig. 1.

We adopt a three-scale UNet configuration, encouraging nonlocal processing at different resolution levels. A single convolutional layer first extracts 32 shallow features from the concatenated image and noise map. These are then passed to the UNet core, where each scale processes features with the same three sequential components: our proposed Nonlocal Feature Matching and Filtering (NLFeMF, detailed in Section 3.2), preceded and succeeded by ConvNeXt blocks (Liu et al., 2022), both without any normalization. In this arrangement (called NL block in Fig. 1), the ConvNeXt blocks, with their $7 \times 7$ depthwise convolutions, provide essential local context extraction and feature refinement around each pixel, complementing the nonlocal processing performed by our feature matching and filtering block.

Downsampling uses 2-strided $2 \times 2$ convolutions and doubles the number of features, whereas we use their transposed equivalent for upsampling and feature halving. Skip connections link down and up blocks in corresponding scales, adding back the residual progressively while decoding. Finally, a single convolutional layer decodes the refined high-resolution features into a denoised image of the same size and channels as the input noisy image.

### 3.2. Nonlocal feature matching and filtering block

The Nonlocal Feature Matching and Filtering (NLFeMF) block is the core component of our proposed network. It implements a learnable variant of the block matching and collaborative filtering paradigm of classical patch-based methods, adapted to operate on learned feature representations rather than image patches. The block operates in three sequential steps: nonlocal feature matching, collaborative filtering, and aggregation. Each of these steps is built of a specific configuration of convolution and deformation layers, designed to imitate the purpose of their classical counterparts. Their detailed composition is pictured in the bottom right of Fig. 1, and explained as follows.

#### 3.2.1. Feature matching

The first step identifies, for each pixel, the $K$ optimal neighboring positions in a surrounding search window, based on their local features, to use afterward in collaborative filtering. Rather than using a hand-crafted similarity metric, we learn the matching process directly. Specifically, a convolutional neural network $\mathfrak{C}_6$—akin to the offset estimation mechanism used by Liang et al. (2022) in their guided deformable attention module for feature alignment between video frames—consisting of six $3 \times 3$ convolutional layers with Leaky ReLU activations between them (with negative slope of 0.1) predicts $K$ two-dimensional offset vectors for each pixel. A scaled hyperbolic tangent activation is used to restrict their values to the defined search window. That is, if $\mathbf{u} : \Omega \to \mathbb{R}^C$ is the feature map of the current scale and $\mathbf{x} = (x, y) \in \Omega$ is a pixel in the image domain, the offsets are computed as

$$\begin{aligned} &\mathbf{o} : \Omega \to [-s, s]^{2K} \\ &\mathbf{o}(\mathbf{x}) = s \tanh(\mathfrak{C}_6(\mathbf{u}; \Theta_o)(\mathbf{x})) \\ &\qquad = (o_1^x(\mathbf{x}), o_1^y(\mathbf{x}), \dots, o_K^x(\mathbf{x}), o_K^y(\mathbf{x})), \end{aligned} \tag{1}$$

where $\Theta_o$ are the learnable parameters of the CNN, and $s$ is the scaling factor that defines the size of the search window. We use $s = 9$ for every scale, so that the search region is always of size $19 \times 19$ independently of $\Omega$.

The predicted offsets are not constrained to integer coordinates, so we apply bilinear interpolation to extract features at subpixel locations. These neighboring features are then stacked on their corresponding pixel position, transforming a feature map with $C$ channels into a grouped representation

$$\begin{aligned} &\mathbf{w} : \Omega \to \mathbb{R}^{C \cdot K} \\ &\mathbf{w}(\mathbf{x}) = \bigoplus_{k=1}^{K} \mathbf{u}(x + o_k^x(\mathbf{x}), y + o_k^y(\mathbf{x})), \end{aligned} \tag{2}$$

being $\oplus$ the concatenation operator. This complete operation is represented by the "Deform" block in Fig. 1.

#### 3.2.2. Collaborative filtering

To filter the stack of neighboring features, we first pass it through a $1 \times 1$ grouped convolutional layer for the $C$ groups. This operation learns a shared linear transformation $\mathfrak{F} : \mathbb{R}^K \to \mathbb{R}^K$ (with parameters $\Theta_f$) that is applied to the stack of neighbors $\mathbf{w}_{c,:}$ sliced at each feature $c$, effectively projecting them into a representation adapted for denoising:

$$\hat{\mathbf{w}}(\mathbf{x}) = \bigoplus_{c=1}^{C} \mathfrak{F}(\mathbf{w}_{c,:}(\mathbf{x}); \Theta_f). \tag{3}$$

Note that grouping is essential so that there is no mixing of possibly-contrasting features, and that only closely resembling matched features are transformed and filtered together. This mirrors classical methods using a linear transform on a block of similar patches to obtain a sparse representation where signal and noise information can be easily separated.

After transformation, we want to suppress the coefficients in the new representation that do not contribute to the image signal information and keep those that do. We do so with a learned modulation map that shrinks the coefficients depending on the local structure of the stacked features. This map $\hat{\mathbf{m}}$ is built by passing the whole non-transformed stack

**Table 2**
Summary of the RAW dataset composition.

| Source | Train | Val | Resolution | Image type |
|---|---|---|---|---|
| ELD | 5 | 1 | $5568 \times 3708$ | Low light |
| SID | 142 | 11 | $4288 \times 2848$ | Low light |
| SIDD | 76 | 0 | Varied (16MP) | Smartphone |
| RawNIND | 106 | 8 | $6048 \times 4024$ | Natural light |
| Nikon | 12 | 1 | $6016 \times 4016$ | Low light |
| CRVD[a] | 10 | 1 | $1920 \times 1080$ | Varied |
| ReCRVD[a] | 109 | 8 | $1920 \times 1080$ | Varied |

[a] Video datasets. Only one frame per scene is used.

to a CNN $\mathfrak{C}_3$ (with parameters $\Theta_m$) composed of three $3 \times 3$ depthwise convolutional layers with ReLU activations and a final sigmoid to get an attenuation coefficient between 0 and 1 for each one of the $C \cdot K$ channels:

$$\begin{aligned} &\hat{\mathbf{m}} : \Omega \to [0,1]^{C \cdot K} \\ &\hat{\mathbf{m}}(\mathbf{x}) = \mathrm{sig}(\mathfrak{C}_3(\mathbf{w}; \Theta_m)(\mathbf{x})). \end{aligned} \tag{4}$$

The depthwise convolutions operate spatially on the stacked features, allowing information from nonlocal neighbors at nearby pixel locations to interact. This effectively expands the receptive field by creating a larger nonlocal context. We multiply the learned modulation map elementwise with the transformed feature stack to perform the filtering: coefficients which contain meaningful texture information (with a multiplier close to 1) are preserved, those that do not (multiplier close to 0) are suppressed, and intermediate values allow for partial attenuation, enabling the network to balance noise reduction with detail preservation in a differentiable manner.

Finally, we apply another transformation to reconstruct the feature stack from the shrunk coefficients. Since the initial groupwise transformation $\mathfrak{F}$ is not guaranteed to be invertible, we learn a separate mapping $\mathfrak{B} : \mathbb{R}^K \to \mathbb{R}^K$ using another $1 \times 1$ grouped convolutional layer, leading to another $CK$-channel representation $\tilde{\mathbf{w}}$ of the stack of filtered features. The filtered stack is thus reconstructed as

$$\tilde{\mathbf{w}}(\mathbf{x}) = \bigoplus_{c=1}^{C} \mathfrak{B}(\hat{\mathbf{m}}_{c,:}(\mathbf{x}) \odot \hat{\mathbf{w}}_{c,:}(\mathbf{x}); \Theta_b), \tag{5}$$

where the subindex notation again slices the modulation map and transformed neighbor stack at feature $c$, $\odot$ is the elementwise multiplication operator, and $\Theta_b$ are the learnable parameters of $\mathfrak{B}$.

### 3.2.3. Aggregation

The final step reduces the $CK$-channel representation back to $C$ channels. Since the whole block of neighbors is filtered simultaneously for every position in the image, some of these neighbors may have multiple denoised estimates. However, their features may have been interpolated at subpixel locations, so aggregating the neighbors back to their position with a simple averaging is an ill-defined operation. Instead, we perform aggregation in feature space using a single $1 \times 1$ convolutional layer $\mathfrak{A} : \mathbb{R}^{C \cdot K} \to \mathbb{R}^C$ that learns to optimally combine the filtered neighbors. The final output of the NLFeMF block is thus given by

$$\hat{\mathbf{u}}(\mathbf{x}) = \mathfrak{A}(\tilde{\mathbf{w}}(\mathbf{x}); \Theta_a), \tag{6}$$

where $\Theta_a$ are the learnable parameters of $\mathfrak{A}$.

## 3.3. RAW database

Since the scarcity of high quality real data may hinder generalization to different sensors, we create our own synthetic noisy images for training. However, we avoid unprocessing (Brooks et al., 2019), as RAW data generated this way retains the irreversible quality loss of the original 8 bit images, which have already undergone quantization, tone mapping and compression. Instead, we curate a dataset of high-quality ground truth RAW images from existing sources, to which we add synthetic Poisson-Gaussian noise. We manually inspect each clean image

**Fig. 2.** Estimated noise level curves of the used camera sensors for different ISO values (labeled to the right of the curves). The standard deviation of the noise is the square root of a linear model with respect to the pixel intensity. Data in the $[0, 1]$ range after black level subtraction.

in SID (Chen et al., 2018), ELD (Wei et al., 2022), SIDD (Abdelhamed et al., 2018), RawNIND (Brummer & Vleeschouwer, 2025), Nikon (Prabhakar et al., 2021), CRVD (Yue et al., 2020) and ReCRVD (Yue et al., 2025), discarding those with residual noise, and retain only Bayer CFA patterns. For the video datasets, we select a single frame per scene to prevent bias from temporal redundancy. This yields 460 images in the training set and 30 for validation. Table 2 summarizes the composition of the curated dataset.

To generate diverse enough noise, we sample the characteristics of seven different sensors at varying ISO levels. We use the noise parameters of five sensors from the SIDD dataset (Samsung S6, iPhone 7, Google Pixel, Nexus 6, LG G4), as well as those from the Sony IMX385 and IMX586 sensors. The parameters from the latter two are specified by Yue et al. (2020) and Wang et al. (2020), respectively. Since the noise level functions embedded in SIDD have been shown to be miscalibrated (Zhang et al., 2021), we reestimate them from the data with the method proposed by Colom and Buades (2013). We measure variance levels for multiple intensity bins for all the images in the dataset, stratifying them by sensor model and ISO level, remove outliers found at high intensity values, and fit a linear curve to the measured points of each group. The slope $a$ of the fitted curve and the intercept $b$ are estimators of the shot and readout noise parameters in a signal-dependent noise model with variance $\sigma^2(x) = ax + b$, where $x$ is the pixel intensity. The curves for the noise standard deviation of the used sensors are shown in Fig. 2. On noise generation during training, we randomly select a sensor and an ISO value (within the available range for the sensor), interpolate the noise curve to that ISO, and generate the noisy data so that each noisy pixel is drawn from the Poisson-Gaussian distribution with those parameters, as

$$x_{\text{noisy}} \sim a\mathcal{P}\left(\frac{x_{\text{true}}}{a}\right) + \mathcal{N}(0, b). \tag{7}$$

This procedure yields diverse RAW training data that spans a realistic range of noise conditions while keeping the image quality encountered in real sensors. Note that the noise standard deviation map during training must be estimated from the shot and readout coefficients using the noisy intensity values rather than the clean ones, so as to prevent the network from retrieving the ground truth signal from the noise map.

## 3.4. Training details

We train the network end-to-end with the $L_1$ loss between denoised and clean images in the RAW domain. On each training step, we crop a single random square of size $128 \times 128$ pixels from the selected 4-channel packed image, augment it with a random transformation from the dihedral group $D_8$, and add to it synthetic noise generated with the procedure explained in Section 3.3. This is done for a total of 10000

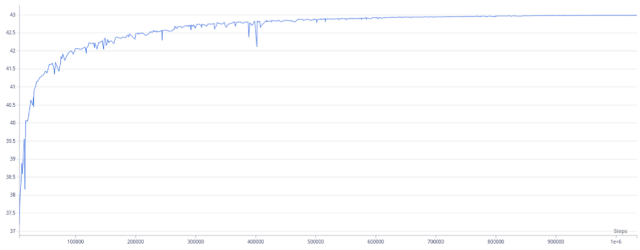

**Fig. 3.** Validation PSNR curve during training on the RAW dataset. The network converges after around 10000 epochs.

epochs with a batch size of 4, amounting to 1.15M iterations. We use Adam (Kingma & Ba, 2017) as optimizer, starting with a learning rate of $10^{-4}$ and gradually reducing it to $5 \cdot 10^{-7}$ with a cosine annealing strategy (Loshchilov & Hutter, 2017). Fig. 3 shows the validation PSNR curve during training, verifying convergence.

## 4. Network analysis

Before evaluating our method on real RAW data, we analyze the proposed architecture in the controlled setting of synthetic Gaussian noise. This simplified scenario allows us to isolate the effects of individual design choices and compare different network configurations more easily. We first describe the minor adaptations required to train the network for this setting, and then study how the number of neighbors, the multiscale structure, and the matching, modulation and aggregation strategies affect denoising performance. Then, to position the proposed nonlocal architecture within the broader landscape of denoising paradigms, we perform quantitative and qualitative comparisons against distinguished classical, CNN-based and transformer-based Gaussian denoisers, as well as perform a comparative analysis on computational complexity. Finally, back in the RAW domain, we study the ability of our network to adapt to different sensors and miscalibrations of the noise level map.

### 4.1. Ablation study

To adapt the proposed network to the Gaussian noise setting, we only modify the input and output channel sizes of its first and last layers, keeping the rest of the architecture as-is: the first layer receives 4-channel data, where the last channel holds the same value of noise standard deviation for each pixel, while the last layer outputs 3-channel RGB images. Following previous works (Liang et al., 2021; Zhang et al., 2022), we train the network on images from the Waterloo Exploration Database (Ma et al., 2017), DIV2K (Agustsson & Timofte, 2017), and Flickr2K (Lim et al., 2017), with grayscale images removed to avoid biases when generating color noise. In total, after splitting the data, we use 7969 images for training and 292 for validation. We add synthetic white Gaussian noise on demand at each step, choosing its standard deviation uniformly at random in the interval $[5, 50]$. As the Gaussian noise dataset is larger than our RAW database, we decrease the number of epochs to 2000 and raise the batch size to 16, reaching around 1M iterations. The chosen optimizer, learning rate, and scheduler are identical to the RAW setting.

All ablation experiments are performed on the CBSD68 test dataset (Martin et al., 2001), to which we add synthetic Gaussian noise with various standard deviations $\sigma \in \{15, 25, 50\}$. For quantitative comparisons, we use color PSNR as the evaluation metric.

#### 4.1.1. Effects of the number of neighbors

To analyze the effect of varying the number of neighbors $K$ used by the network, we build and train a simplified single-scale network containing an individual NL block plus the $3 \times 3$ head and tail convolutions. As shown in Table 3, increasing $K$ leads to higher PSNR values at the cost of additional network complexity. The parameter increase is consequence of the filtering step needing to learn a set of linear filters for the stack of $K$ matched neighbors. The gains in PSNR diminish for large $K$, as the additional neighbors become progressively more redundant. These results go in accordance with classical patch-matching methods, where increasingly dissimilar patches contribute only marginally to the filtering.

**Table 3**
Effect of the number of neighbors $K$. PSNR measured on CBSD68 for AWGN denoising. A slash / separates neighbors from different scales, from higher to lower resolution. Rows are grouped as: 1 scale, 3 scales.

| $K$ | Params (M) | PSNR (dB) | | |
|---|---|---|---|---|
| | | $\sigma = 15$ | $\sigma = 25$ | $\sigma = 50$ |
| 5 | 0.139 | 33.79 | 31.12 | 27.81 |
| 9 | 0.322 | 33.90 | 31.23 | 27.95 |
| 15 | 0.786 | 33.98 | 31.32 | 28.06 |
| 25 | 2.1 | 34.03 | 31.37 | 28.11 |
| 35 | 4.0 | 34.07 | 31.42 | 28.16 |
| 49 | 7.7 | 34.11 | 31.45 | 28.18 |
| 63 | 12.6 | 34.14 | 31.48 | 28.22 |
| 81 | 20.8 | 34.15 | 31.50 | 28.24 |
| 5/5/5 | 3.5 | 34.23 | 31.61 | 28.41 |
| 15/7/5 | 5.4 | 34.30 | 31.67 | 28.47 |
| 15/9/7 | 7.5 | 34.31 | 31.69 | 28.50 |
| 25/15/9 | 15.3 | 34.34 | 31.72 | 28.53 |

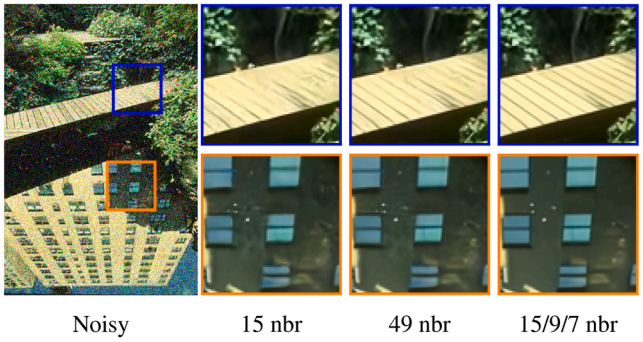


**Fig. 4.** Visual comparison between different values of $K$ at different scales. A slash separates neighbors from different scales. Noisy image is corrupted with Gaussian noise with $\sigma = 50$.

Extending the architecture to multiple scales yields a substantial improvement in PSNR, as it can also be seen in Table 3. This improvement is noticeable even when the number of neighbors at each scale is low: since spatial resolution decreases and feature dimensionality increases at coarser scales, fewer neighbors are required at those levels to maintain denoising quality.

Fig. 4 shows a visual comparison between different arrangements, using a single scale with $K = 15$ and $K = 49$, and three scales with 15, 9 and 7 neighbors, respectively. Looking at the blue patch, it is clear that increasing the number of neighbors on a single scale helps reconstruct structured patterns that repeat in the image. Increasing the number of scales further improves the reconstruction quality, which is especially noticeable in fine textures like the lines on the bridge and the vegetation behind it. Note that the latter two network configurations use approximately the same number of parameters (7.7M and 7.5M). Even if each scale individually uses fewer neighbors than in the single scale case, the neighbors at lower scales retrieve spatial relations inside the image more effectively, which allows the network to discern noise from texture more precisely. The use of multiple scales also helps remove low-frequency artifacts that appear at high noise levels, like the one in the orange patch: the hierarchical structure allows for a progressive and more robust neighbor selection, which translates to higher image quality.

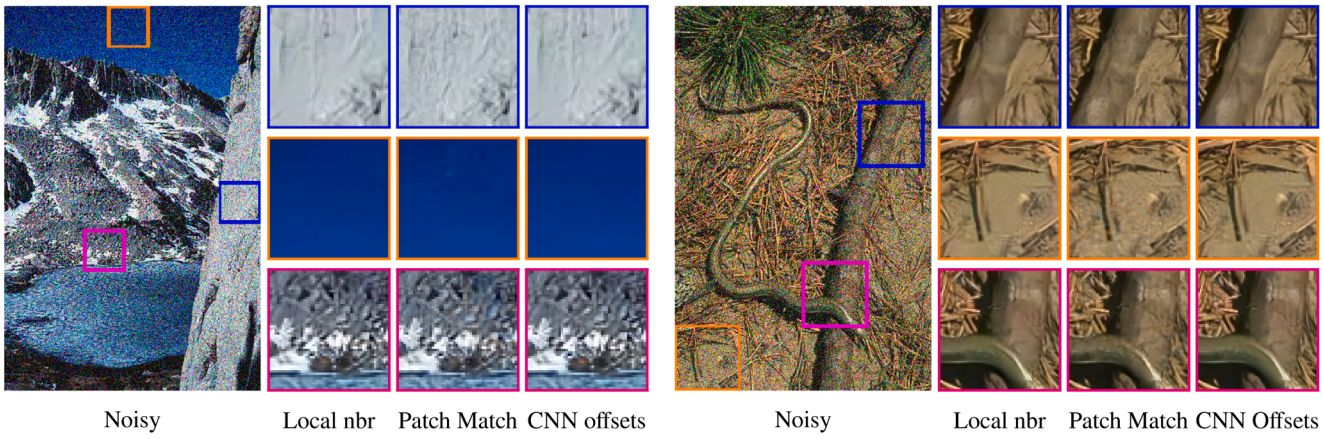


**Fig. 5.** Visual comparison between different feature matching strategies on two images from the CBSD68 dataset. A single scale with a NL block with $K = 15$ and 6 offset layers is used for denoising. Noisy images are corrupted with Gaussian noise with $\sigma = 50$.

**Table 4**
Effect of the feature matching strategy. A single scale with a NL block with $K = 15$ is used. PSNR measured on CBSD68 for AWGN denoising.

| Matching strategy | PSNR (dB) | | |
|---|---|---|---|
| | $\sigma = 15$ | $\sigma = 25$ | $\sigma = 50$ |
| Local neighbors | 33.92 | 31.24 | 27.93 |
| Differentiable Patch Match | 33.76 | 31.09 | 27.63 |
| Offset CNN (2 layers) | 33.95 | 31.28 | 28.01 |
| Offset CNN (4 layers) | 33.96 | 31.30 | 28.03 |
| Offset CNN (6 layers) | 33.98 | 31.32 | 28.06 |
| Offset CNN (10 layers) | 33.99 | 31.33 | 28.05 |

#### 4.1.2. Feature matching strategy

We compare our learned convolutional neighbor search against two alternatives: using a fixed local window around each pixel as the neighbor set, and the differentiable Patch Match algorithm proposed by Cherel et al. (2024). We do so within a single scale, with $K = 15$ neighbors, or an equivalent $3 \times 5$ window for the local baseline, visualizing the selected neighbors for each strategy. We also vary the number of convolutional layers in the offset CNN to study its effect on the matching quality.

Table 4 shows that the learned CNN-based feature matching outperforms the local window baseline. The depth of the CNN is not critical, as even a shallow network with 2 layers is enough to surpass the baseline. At the chosen depth of 6 convolutions, the improvement over it becomes moderate, increasing with the noise level. Adding more layers does not yield further significant gains, suggesting that the network has enough capacity to learn a good matching function given the defined search window size. On the other hand, the differentiable Patch Match performs notably worse, particularly at high noise levels. This is illustrated in Fig. 5. While Patch Match does a good job reconstructing textured areas like the wall and the log in the blue patches, it produces low-frequency artifacts in flat regions (orange patches). Results using this matching strategy can also suffer from color bleed. This can be seen in the magenta patches, where the gray rocks on the mountain have a blue tint, and the green snake adopts the browner color of the log. We attribute these limitations to the difficulty of matching noisy features in a high-dimensional space, and the consequent proliferation of incorrectly computed shifts during the propagation step of the algorithm. Compared to the local baseline, the 6-layer CNN-based matching leads to better reconstruction of texture in all cases.

The differences in how the matching strategies select neighbors can be seen in Fig. 6 for a given query point in two textured regions of an image of the CBSD68 dataset. Local neighbors form a tight cluster around the query, restricting support to a small spatial window. Patch Match retrieves a more scattered set of candidates, as the algorithm is able to propagate matches from distant locations. However, some of the retrieved neighbors seem to be mismatched due to the noise when looking at them in the clean image (especially with the bottom query, where most neighbors fall outside the shown patch), and self-similar textures closer to the query are not fully exploited. The neighbors selected by our CNN-based matching distribute adaptively along the dominant image structures, even through noise. For example, in the blue patch, the neighbors follow the tubes on the coral, forming a coherent straight line. In the orange patch, the same happens at a different angle; in this case,

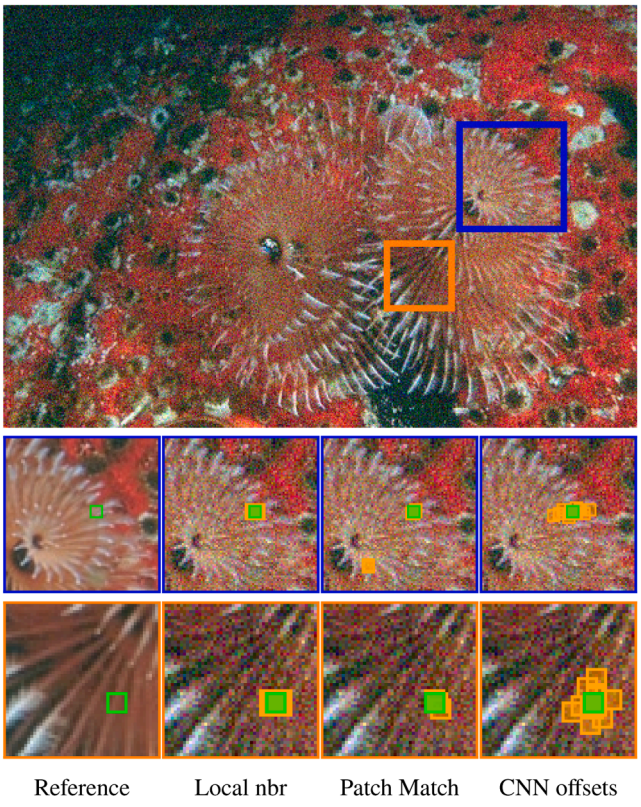


**Fig. 6.** Neighbors ($K = 15$) selected by different feature matching strategies, painted as orange colored patches. Query position is painted green and shown over the ground truth image. Some patches selected by Patch Match fall outside the crops. Noisy image corrupted with Gaussian noise with $\sigma = 25$.

**Table 5**
Effect of the collaborative filtering and aggregation strategies. A single scale with a NL block with $K = 15$ is used. PSNR measured on CBSD68 for AWGN denoising.

| Modulation | Aggregation | PSNR (dB) | | |
|---|---|---|---|---|
| | | $\sigma = 15$ | $\sigma = 25$ | $\sigma = 50$ |
| Soft threshold | Learned | 33.68 | 31.00 | 27.68 |
| Learned | Indexing | 33.85 | 31.18 | 27.90 |
| Learned | Learned | 33.98 | 31.32 | 28.06 |

two tubes are within the search window around the query point, and the network selects neighbors along both of them. The texture in this latter patch is subtler and obfuscated by noise, yet the network is able to find suitable matches.

#### 4.1.3. Collaborative filtering and aggregation methods

We also study the effect of the collaborative filtering and aggregation strategies on denoising performance. Again, we use a single scale with $K = 15$ neighbors, comparing two alternatives to each of the steps in the proposed nonlocal block. For the collaborative filtering, we compare the learned modulation map against a soft thresholding function. As the choice of threshold depends on the learned domain of the transformed coefficients and needs to be differentiable, we choose a logistic function with a learnable midpoint $m$ and growth rate $g$, so that the map in equation (4) becomes

$$\hat{\mathbf{m}}(\mathbf{x}; g, m) = \frac{1}{1 + e^{-g(\tilde{\mathbf{w}}(\mathbf{x}) - m)}}. \tag{8}$$

For the aggregation step, we compare the learned $1 \times 1$ convolutional layer in (6) against a simple indexing operation that selects the first neighbor in the stack of filtered features:

$$\hat{\mathbf{u}}(\mathbf{x}) = \tilde{\mathbf{w}}_{:,1}(\mathbf{x}), \tag{9}$$

where the subscript indicates the slicing of the first index.

Table 5 shows that the learned modulation and aggregation strategies outperform their alternatives, particularly at high noise levels. The benefits of using a spatially-dependent modulation map over a classical soft thresholding is especially noticeable, with a PSNR advantage ranging from 0.3 to 0.4 dB in PSNR in CBSD68. The learned convolutional aggregation also improves the results over just keeping the first neighbor, with a minimal overhead in complexity.

### 4.2. Comparison with Gaussian denoisers

For completeness and to facilitate comparison with the broader denoising literature, we measure our method against the state of the art in AWGN denoising. We compare it against denoisers spanning classical, CNN-based and transformer-based paradigms, as well as other nonlocal strategies closely related to our approach: BM3D (Dabov et al., 2007) for the former; DnCNN (Zhang et al., 2017a), FFDNet (Zhang et al., 2018) and DRUNet (Zhang et al., 2022) for CNNs; Restormer (Zamir et al., 2022), CTNet (Tian et al., 2024b), Condformer (Huang & Huang, 2025) and DSCA-Former (Hu et al., 2026) for transformers; and NLNet (Lefkimmiatis, 2017), NLRN (Liu et al., 2018), N3Net (Plötz & Roth, 2018) and LNSNet (Meng et al., 2024) for the latter. For all methods, if available, we use their officially released parameter checkpoints trained on the broadest available noise range. Quantitative results are reported on CBSD68 (Martin et al., 2001), Kodak (Franzen, 1999) and McMaster (Zhang et al., 2011) for $\sigma \in \{15, 25, 50\}$.

Table 6 summarizes the obtained PSNR and SSIM values. Our three-scale configuration with 15/9/7 neighbors matches or exceeds DRUNet on all datasets, despite using less than one-fourth of the parameters (7.5M vs. 32.6M). Using fewer neighbors in the coarsest scales to a 15/7/5 configuration reduces the number of parameters even further (5.4M) while maintaining competitive results: at a similar parameter count to this smaller configuration, the nonlocal network LNSNet (6.5M parameters) performs similarly at low noise levels, but reports lower metrics at $\sigma = 25$ and $\sigma = 50$, most notably on the Kodak dataset. The larger 25/15/9 variant performs comparably to CTNet and DSCA-Former at low noise levels and surpasses them at $\sigma = 50$, despite having significantly fewer parameters (15.3M vs. 49M and 65.1M). It also offers lower network complexity than Restormer and Condformer (26.1M and 27.0M parameters), against which the margin remains small. On the opposite end of network size, the lightweight single-scale version of our network (786k parameters) consistently outperforms BM3D, DnCNN, FFDNet, NLNet and NLRN. Against N3Net, another nonlocal network of similar size, it also achieves better or comparable results in CBSD68 and Kodak, with a wide gap in McMaster.

Fig. 7 shows a visual example comparing the single-scale variant with 15 neighbors against the lighter methods. Out of the visualized approaches, FFDNet and ours are the ones with fewer denoising artifacts. Every tested method, including ours, struggles to recover the fine thread pattern in the blue patch. NLNet and N3Net introduce some slight gray banding at the center of the thread roll, where the thread seems thicker, while FFDNet and our method return a smooth reconstruction. BM3D also smoothes the high frequency noise and texture, but it also introduces low frequency noise with a vertical pattern following the shadow at the center of the roll. Next, the orange patch shows a heavily patterned area with flat texture. FFDNet leaves residual noise, while N3Net oversmoothes the tie, blending the polka dots together. The other nonlocal methods denoise the area correctly. Overall, our result is sharper than that of BM3D and NLNet; however, some dots appear slightly deformed. This is a limitation of our single-scale variant, caused by imprecise offsets computed on noisy features.

The complete three-scale 25/15/9 variant is compared against DRUNet, Restormer, Condformer and CTNet on images of the Urban100 dataset (Huang et al., 2015) in Fig. 8. All methods yield high quality results, with only subtle differences. CTNet and our method produce slightly better reconstructions of thin straight lines, as it can be seen in the center and bottom right of the top blue patch, where the other three methods fade them too much. Restormer tends to smooth out flat regions with low-frequency grain texture, like the shaded region at the left of the balustrade in the top orange patch. The other four methods do not exhibit this issue as notably. The patches of the bottom image highlight how Restormer, Condformer and CTNet can generate hallucinated textures when trying to reconstruct fine details. In the blue patch, Restormer and Condformer (both with a similar architecture) erroneously extend the step of the staircase into the wall, creating a blending effect. DRUNet and our method correctly filter the texture on the wall; CTNet manages to further recover some of the stonelike pattern on it, but causes a ringing artifact around the step. CTNet also fabricates a curved line between steps of the staircase in the orange patch, which clearly should not be present. Overall, our method achieves a visually pleasant reconstruction that completely eliminates noise while precisely recovering fine details and avoiding hallucinations.

Finally, to better understand the behavior of the networks, we visualize their effective receptive field (Luo et al., 2016) by backpropagating the gradient of a single query pixel in the output with respect to the input image. We compare the resulting heatmap of our method against that of DRUNet, Restormer and N3Net (representing different learning paradigms) in Fig. 9, choosing a pixel in the region of the bottom blue patch in Fig. 8 where Restormer hallucinates. Despite Restormer's receptive field being global due to its use of transposed self-attention, the most influential pixels concentrate around the query point and follow the step of the staircase, which may explain the hallucination of the step into the wall. DRUNet better follows the wall, although in a quasi-isotropic pattern where dissimilar pixels on the stair-

**Table 6**
Average PSNR (dB, top of cell) and SSIM (bottom of cell) of various methods on the CBSD68, Kodak and McMaster test datasets for AWGN denoising. The compared methods use the same set of parameter weights across all noise standard deviations when such checkpoint is available. Methods are grouped as: classical, CNN-based, transformer-based, nonlocal neural networks, and ours.

| Method | Params (M) | $\sigma = 15$ | | | $\sigma = 25$ | | | $\sigma = 50$ | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | CBSD68 | Kodak | McM | CBSD68 | Kodak | McM | CBSD68 | Kodak | McM |
| BM3D | – | 33.46<br>0.9287 | 34.34<br>0.9230 | 33.99<br>0.9375 | 30.68<br>0.8773 | 31.74<br>0.8759 | 31.60<br>0.9012 | 27.21<br>0.7809 | 28.42<br>0.7889 | 28.29<br>0.8258 |
| DnCNN | 0.854 | 33.90<br>0.9290 | 34.60<br>0.9209 | 33.45<br>0.9035 | 31.24<br>0.8830 | 32.14<br>0.8775 | 31.52<br>0.8694 | 27.95<br>0.7896 | 28.95<br>0.7917 | 28.62<br>0.7986 |
| FFDNet | 0.854 | 33.87<br>0.9290 | 34.63<br>0.9292 | 34.66<br>0.9433 | 31.21<br>0.8910 | 32.13<br>0.8887 | 32.35<br>0.9131 | 27.96<br>0.8034 | 28.98<br>0.8097 | 29.18<br>0.8508 |
| DRUNet | 32.6 | 34.30<br>0.9391 | 35.31<br>0.9348 | 35.40<br>0.9503 | 31.69<br>0.8983 | 32.89<br>0.8979 | 33.14<br>0.9232 | 28.51<br>0.8173 | 29.86<br>0.8275 | 30.08<br>0.8690 |
| Restormer | 26.1 | 34.39<br>0.9400 | 35.44<br>0.9359 | 35.55<br>0.9511 | 31.78<br>0.8998 | 33.02<br>0.8997 | 33.31<br>0.9247 | 28.59<br>0.8201 | 30.00<br>0.8301 | 30.29<br>0.8724 |
| Condformer | 27.0 | 34.39<br>0.9405 | 35.40<br>0.9361 | 35.52<br>0.9512 | 31.79<br>0.9011 | 32.99<br>0.9005 | 33.30<br>0.9251 | 28.63<br>0.8229 | 29.98<br>0.8320 | 30.30<br>0.8732 |
| CTNet | 49.0 | 34.32<br>0.9392 | 35.24<br>0.9348 | 35.46<br>0.9501 | 31.68<br>0.8980 | 32.79<br>0.8974 | 33.17<br>0.9225 | 28.41<br>0.8140 | 29.65<br>0.8232 | 30.00<br>0.8649 |
| DSCA-Former[a] | 65.1 | 34.35<br>– | 35.42<br>– | 35.58<br>– | 31.72<br>– | 32.87<br>– | 33.28<br>– | 28.48<br>– | 29.67<br>– | 30.21<br>– |
| NLNet | 0.025 | 33.81<br>0.9336 | 34.65<br>0.9275 | 34.10<br>0.9390 | 31.10<br>0.8878 | 32.10<br>0.8848 | 31.72<br>0.9043 | 27.77<br>0.7947 | 28.85<br>0.7994 | 28.47<br>0.8322 |
| NLRN[b] | 0.359 | 31.83<br>0.9143 | 33.05<br>0.9121 | 33.98<br>0.9385 | 29.37<br>0.8596 | 30.70<br>0.8634 | 31.57<br>0.9046 | 26.45<br>0.7576 | 27.82<br>0.7744 | 28.40<br>0.8352 |
| N3Net[b] | 0.708 | 33.96<br>0.9360 | 34.69<br>0.9290 | 33.54<br>0.9383 | 31.35<br>0.8932 | 32.30<br>0.8890 | 31.63<br>0.9078 | 28.12<br>0.8064 | 29.21<br>0.8103 | 28.84<br>0.8452 |
| LNSNet[a] | 6.5 | 34.30<br>0.9337 | 35.18<br>0.9274 | 35.40<br>0.9315 | 31.64<br>0.8900 | 32.70<br>0.8878 | 33.07<br>0.9004 | 28.40<br>0.8043 | 29.57<br>0.8129 | 29.93<br>0.8398 |
| Ours (15 nbr) | 0.786 | 33.98<br>0.9357 | 34.88<br>0.9300 | 34.84<br>0.9452 | 31.32<br>0.8910 | 32.38<br>0.8883 | 32.52<br>0.9145 | 28.06<br>0.8002 | 29.21<br>0.8059 | 29.31<br>0.8500 |
| Ours (15/7/5) | 5.4 | 34.30<br>0.9391 | 35.30<br>0.9346 | 35.37<br>0.9499 | 31.67<br>0.8980 | 32.87<br>0.8973 | 33.11<br>0.9227 | 28.47<br>0.8160 | 29.82<br>0.8257 | 30.02<br>0.8675 |
| Ours (15/9/7) | 7.5 | 34.31<br>0.9392 | 35.33<br>0.9348 | 35.40<br>0.9502 | 31.69<br>0.8982 | 32.90<br>0.8977 | 33.14<br>0.9230 | 28.50<br>0.8164 | 29.85<br>0.8265 | 30.07<br>0.8683 |
| Ours (25/15/9) | 15.3 | 34.34<br>0.9394 | 35.37<br>0.9352 | 35.45<br>0.9505 | 31.72<br>0.8986 | 32.94<br>0.8983 | 33.19<br>0.9235 | 28.53<br>0.8172 | 29.90<br>0.8275 | 30.12<br>0.8691 |

[a] No public code available for reproduction. Results taken from the original papers.
[b] Original checkpoints only available for grayscale denoising. Results obtained after retraining on color images.

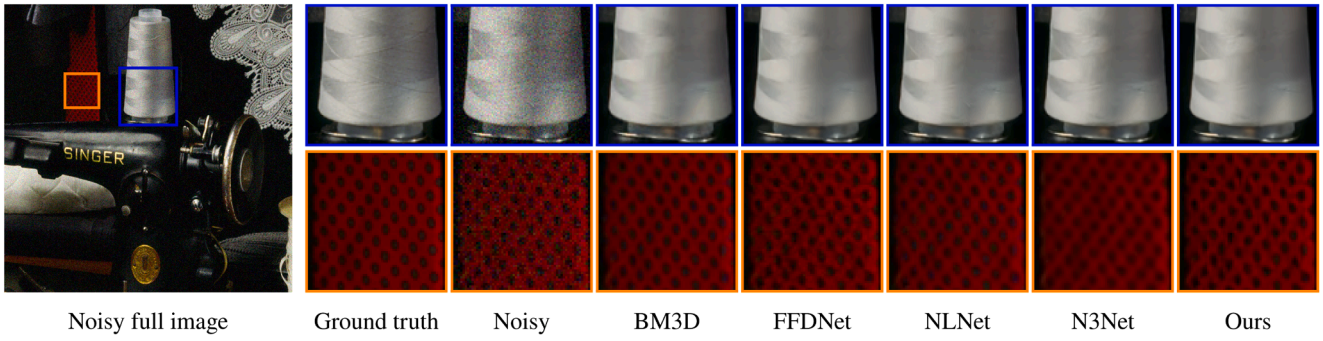


**Fig. 7.** Visual comparison between our lighter single-scale variant with 15 neighbors, BM3D, FFDNet, NLNet and N3Net. The image is corrupted with Gaussian noise with $\sigma = 15$.

case still have some influence. Note its difference with N3Net, which has a smaller support, but with a marked dominant direction along the wall. Our method, to conclude, has a similar spatial reach to DRUNet (with one fewer scale), but exhibits a clearer directional behavior, combining wide coverage with the structure awareness characteristic of nonlocal architectures.

### 4.3. Complexity analysis

In order to assess the computational complexity of our method, we measure the number of floating point operations (FLOPs), the memory footprint and the inference time for a single forward pass on an image of size $256 \times 256$. We compare the profiling results against the same set of

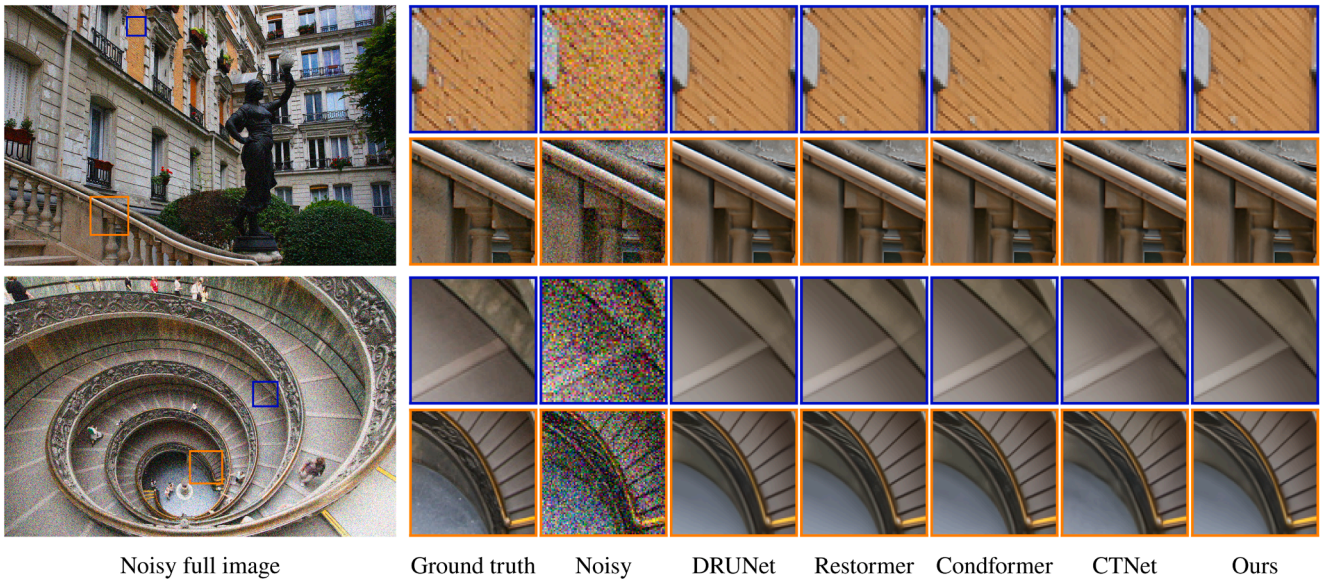


**Fig. 8.** Visual comparison between DRUNet, Restormer, Condformer, CTNet and our method with 25/15/9 neighbors. The top image is corrupted with Gaussian noise with $\sigma = 25$, while the bottom one has $\sigma = 50$.

**Fig. 9.** Effective receptive field heatmap of different denoising methods (CNN, Transformer, nonlocal CNN and our 25/15/9 variant) around a query point (center of crop).

**Table 7**
Computational resources used to denoise an image of size $256 \times 256$ pixels. Methods grouped as in Table 6.

| Method | GFLOPs | Mem. (MiB) | Time (ms) |
|---|---|---|---|
| BM3D[a] | 5.82 | 197.30 | 3557.40 |
| DnCNN | 43.79 | 35.70 | 10.05 |
| FFDNet | 27.94 | 29.53 | 2.16 |
| DRUNet | 277.55 | 228.00 | 10.21 |
| Restormer | 309.76 | 676.67 | 80.07 |
| Condformer | 310.37 | 679.90 | 88.75 |
| NLNet | 11.41 | 163.84 | 93.44 |
| N3Net | 246.34 | 1093.71 | 41.22 |
| LNSNet[b] | 484.08 | 2130.00 | 614.00 |
| Ours (15 nbr) | 103.04 | 643.63 | 8.59 |
| Ours (15/7/5) | 279.06 | 676.32 | 23.83 |
| Ours (15/9/7) | 315.92 | 685.12 | 25.29 |
| Ours (25/15/9) | 782.67 | 1118.94 | 46.81 |

[a] Values measured on CPU.
[b] No code available. Values taken from the original paper.

methods in Section 4.2, with the exception of NLRN and CTNet (which employ patch-based inference and are thus not directly comparable with the rest) and DSCA-Former (which has no public code available nor reported values in the paper).

In Table 7, we share the measured values on a single NVIDIA A6000 GPU, using the official implementations of each method when possible. Our multi-scale variants occupy a middle ground between CNNs and transformers of similar network size in both quality and speed: they run slower than DRUNet, but significantly faster than Restormer and Condformer while achieving competitive denoising performance. The runtime advantage relative to the FLOP count stems from the architectural design of our nonlocal block. Using a single block per scale limits the network depth, which reduces the number of sequential operations and allows for an efficient use of GPU parallelization. Compared to other nonlocal networks, our 15/7/5 and 15/9/7 variants use a fraction of the runtime of NLNet, N3Net and LNSNet, while achieving better results. The single-scale variant with 15 neighbors runs at an average of 8.59 ms—faster than DnCNN and all other nonlocal networks of similar size, and in the same order of magnitude as FFDNet, while outperforming all of them in image quality. Concerning memory usage, the 15/7/5 and 15/9/7 variants require around 680 MiB of VRAM, similar to transformer-based models. Increasing the number of neighbors to 25/15/9 increases the number to 1.1 GiB, a footprint similar to that of N3Net and half of LNSNet. This memory cost is mainly due to storing the full feature stack associated to each neighbor at every pixel position, and so it primarily increases with the number of neighbors in the finest scale. A patch-based inference strategy could reduce it at the expense of a longer runtime. We note that the reported measurements may depend on many factors, including the implementation language (Matlab for NLNet, C++ for BM3D on CPU, Python with PyTorch for the rest) and code optimization, and are not representative of deployment on dedicated hardware. Rather, they should be interpreted as a rough yet meaningful relative comparison between methods.

### 4.4. Generalization and noise map analysis

To evaluate the ability of our method to handle a variety of noise profiles, we study the two key aspects that enable sensor-agnostic denoising: sensor model variety on training, and the input noise map. We do so on the single-scale network variant with 15 neighbors, which we train on the RAW dataset described in Section 3.3.

First, we test the generalization of the network to sensors outside the training set by training it two separate ways: one while holding out the IMX586 from the sensor pool, and another using only that same sensor's noise model. On the validation set, we add synthetic noise from the IMX586 profile with various ISO settings, and compare the denoising performance of both models. Table 8 shows the results, where the values are measured in the sRGB domain after applying a simple pro-

**Table 8**
Denoising performance (single scale, $K = 15$) on the validation set with synthetic noise from the IMX586 sensor, when the network is trained without it or with only that sensor.

| ISO | No IMX586 | | Only IMX586 | |
|---|---|---|---|---|
| | PSNR (dB) | SSIM | PSNR (dB) | SSIM |
| 400 | 39.82 | 0.9706 | 39.63 | 0.9700 |
| 800 | 38.66 | 0.9632 | 38.54 | 0.9628 |
| 1600 | 37.31 | 0.9531 | 37.28 | 0.9531 |
| 3200 | 35.75 | 0.9396 | 35.79 | 0.9400 |
| 6400 | 33.89 | 0.9219 | 33.99 | 0.9225 |

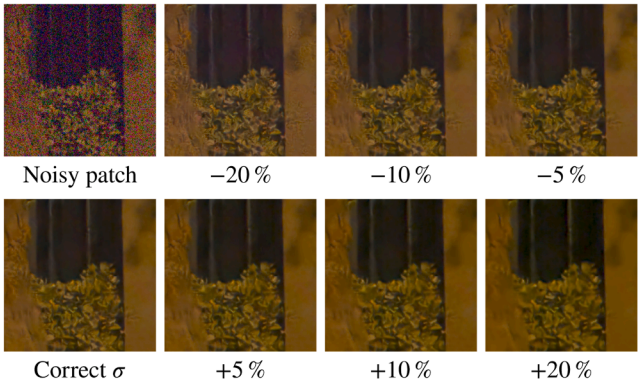


**Fig. 10.** Effect of under- (top) or overestimating (bottom) the noise level by a given factor. Image corrupted with noise from the IMX586 sensor model with ISO 6400, and denoised with our single-scale variant with 15 neighbors.

**Table 9**
Denoising performance (single scale, $K = 15$) when the noise map is inaccurate by $\epsilon$ %. PSNR (dB) and SSIM measured on the validation set with synthetic noise from the IMX586 sensor.

| ISO | $\epsilon$ | Underestimation | | Overestimation | |
|---|---|---|---|---|---|
| | | PSNR | SSIM | PSNR | SSIM |
| 400 | 0 % | 39.87 | 0.9706 | 39.87 | 0.9706 |
| | 5 % | 39.82 | 0.9707 | 39.85 | 0.9702 |
| | 10 % | 39.70 | 0.9703 | 39.79 | 0.9696 |
| | 20 % | 39.18 | 0.9669 | 39.55 | 0.9680 |
| 800 | 0 % | 38.71 | 0.9632 | 38.71 | 0.9632 |
| | 5 % | 38.64 | 0.9633 | 38.69 | 0.9626 |
| | 10 % | 38.47 | 0.9626 | 38.59 | 0.9617 |
| | 20 % | 37.73 | 0.9563 | 38.23 | 0.9594 |
| 1600 | 0 % | 37.39 | 0.9532 | 37.39 | 0.9532 |
| | 5 % | 37.29 | 0.9533 | 37.34 | 0.9523 |
| | 10 % | 37.04 | 0.9521 | 37.17 | 0.9510 |
| | 20 % | 36.00 | 0.9404 | 36.58 | 0.9475 |
| 3200 | 0 % | 35.86 | 0.9398 | 35.86 | 0.9398 |
| | 5 % | 35.72 | 0.9399 | 35.72 | 0.9386 |
| | 10 % | 35.35 | 0.9380 | 35.35 | 0.9365 |
| | 20 % | 33.94 | 0.9179 | 34.28 | 0.9293 |
| 6400 | 0 % | 34.02 | 0.9222 | 34.02 | 0.9222 |
| | 5 % | 33.86 | 0.9222 | 33.65 | 0.9201 |
| | 10 % | 33.29 | 0.9191 | 32.93 | 0.9150 |
| | 20 % | 31.41 | 0.8881 | 31.12 | 0.8974 |

cessing pipeline. It can be observed that both networks achieve similar PSNR and SSIM values, with the biggest disparities appearing at the extremes of the ISO range. These differences are kept within 0.2 dB of PSNR and 0.0006 points of SSIM, and are reduced to less than 0.04 dB and no SSIM difference for intermediate ISO values. Perhaps counterintuitively, the model trained without the IMX586 sensor obtains better results at ISO 400 and 800; this is likely due to the remaining sensors spanning a range of noise characteristics that better cover lower noise levels than the IMX586 alone. Overall, these results corroborate that the diversity of noise profiles in the training set is effective in enabling the network to generalize to unseen sensors.

Next, we analyze the effect of inaccurate noise maps on denoising performance. To do so, we run the network (trained with all sensors) on the same IMX586-corrupted images as before, but artificially scale the input noise map by a factor of $1 \pm \epsilon$, with $\epsilon \in \{0, 0.05, 0.1, 0.2\}$ to simulate underestimation and overestimation of the noise level. The resulting effect is exemplified in Fig. 10, where it can be seen that the filtering strength of the network increases with the noise map scaling factor. Table 9 reports the PSNR and SSIM for different scaling values. As expected, deviations from the correct noise map degrade performance. Underestimation is the more harmful direction, as it hampers the network from fully removing noise, causing it to leave residual artifacts if the map is too inaccurate. The opposite direction results in a smoother image with some loss of detail. This asymmetry is reflected in the PSNR values for ISO 400 to 3200, where overestimating consistently outperforms underestimating the noise by the same amount. The trend shifts in ISO 6400, though the visual behavior remains the same, as seen in Fig. 10. SSIM shows the opposite pattern for moderate deviations ($\epsilon \leqslant 0.1$), likely because the metric favors structured noise in textured areas over smoothed details. At $\epsilon = 0.2$, both metrics drop sharply across all ISO values and directions. Still, the network remains robust to moderate deviations up to $\pm 10$ %, well within the margin of error of modern noise estimators (Idoughi et al., 2025).

**Table 10**
Results on the Darmstadt Noise Dataset benchmark for RAW image denoising. The values are measured in the sRGB domain after applying the benchmark's own processing pipeline.

| Method | PSNR (dB) | SSIM |
|---|---|---|
| BM3D | 37.78 | 0.9308 |
| DnCNN | 38.08 | 0.9357 |
| N3Net | 38.32 | 0.9384 |
| Ours (15 nbr) | 39.99 | 0.9594 |
| UPI | 40.35 | 0.9641 |
| CycleISP | 40.50 | 0.9655 |
| PseudoISP | 40.36 | 0.9606 |
| DualDn | 40.70 | 0.9635 |
| Ours (25/15/9 nbr) | 40.63 | 0.9644 |

## 5. Experimental results on real RAW data

We finally conduct experiments to assess the performance of our method on real RAW data. We evaluate it on two complementary bases: the Darmstadt Noise Dataset (DND) (Plötz & Roth, 2017) for standardized testing, and the in-the-wild smartphone captures provided by Li et al. (2024b) for visual assessment of generalization across different sensors. None of the sensors in these benchmarks coincide with the ones used for training our network: DND contains captures from a Sony A7R, an Olympus OM-D E-M10, a Sony RX100 IV, and a Huawei Nexus 6P, while the latter dataset's scenes are taken with three additional consumer smartphone models whose specific sensors are not disclosed. Together, they provide a rigorous test of generalization to real unseen sensor hardware.

### 5.1. Results on DND

We compare our 25/15/9 network variant against the top-performing methods on the RAW denoising benchmark shared in the DND web platform: UPI (Brooks et al., 2019), CycleISP (Zamir et al., 2020), PseudoISP (Cao et al., 2024), and DualDn (Li et al., 2024b). Note

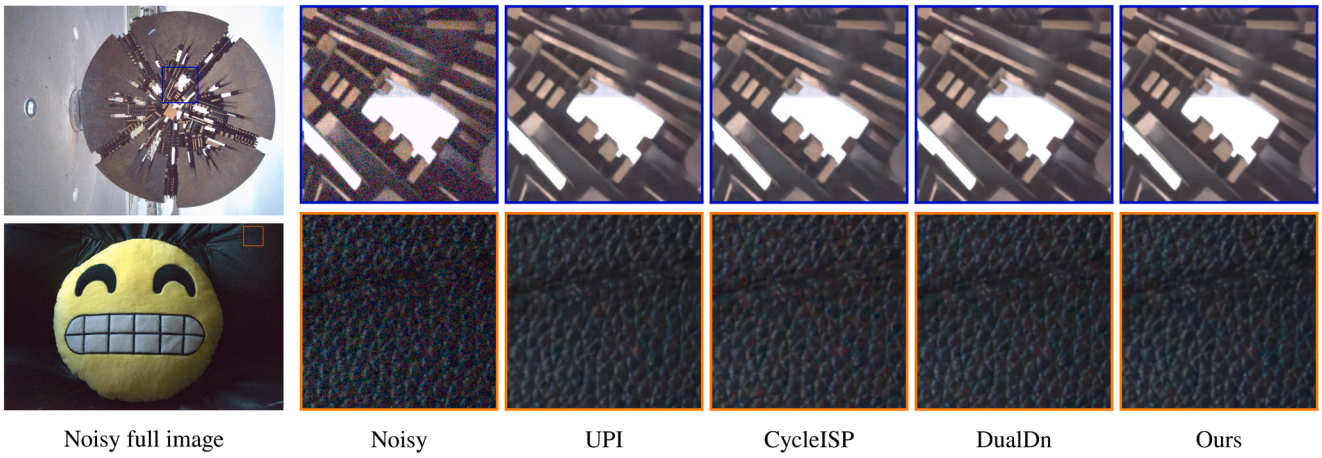


**Fig. 11.** Visual comparison against state-of-the-art RAW denoising methods on the DND dataset. Denoised images are retrieved from the official benchmarking platform after being processed with an internal pipeline.

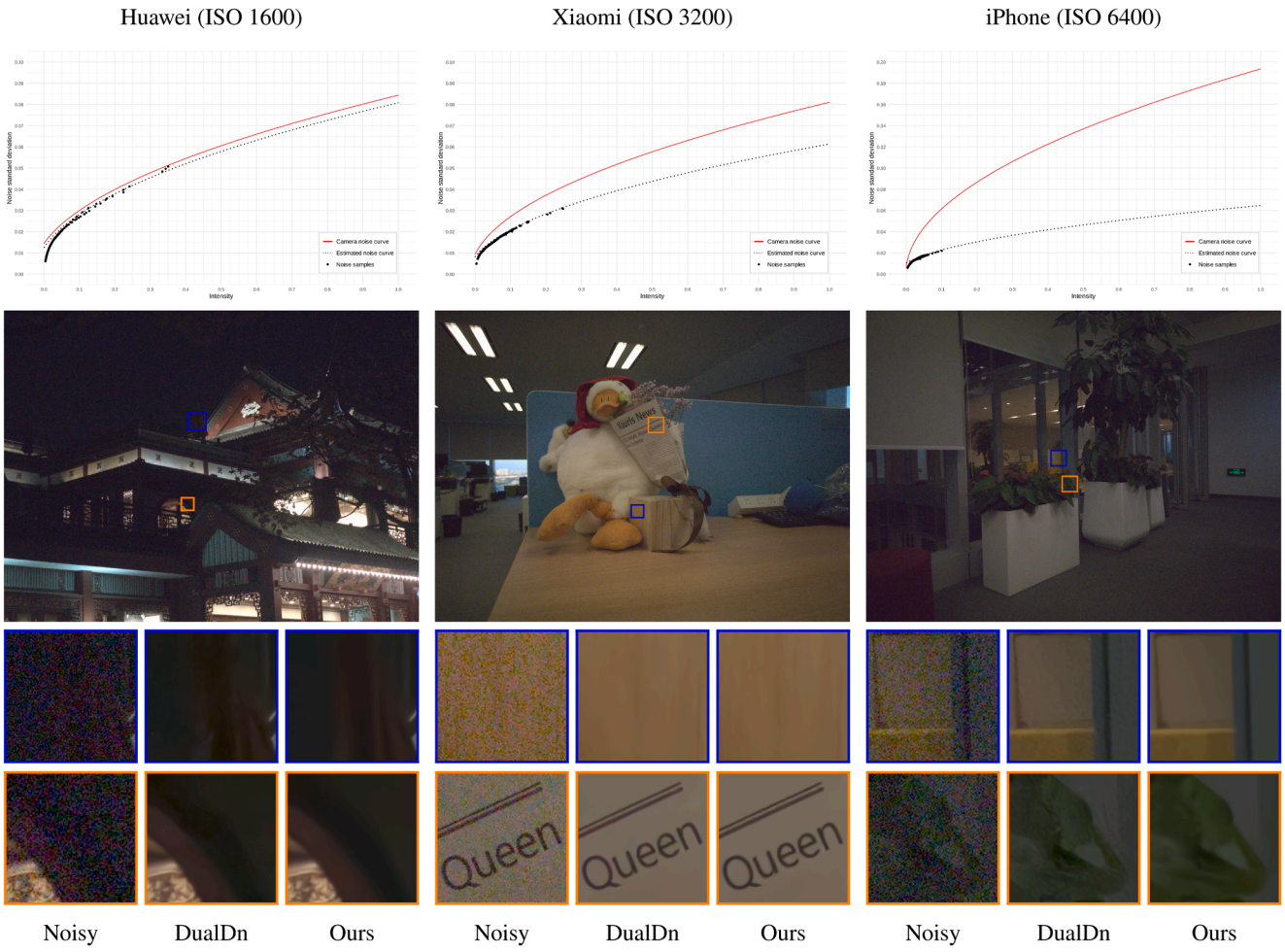


**Fig. 12.** Visual comparison on in-the-wild smartphone photographs. Each column shows an image captured with a different device and ISO. First row: noise curve of the image, as embedded in the file metadata (red) and as estimated by us (dotted black). Second row and below: noisy image, and patches denoised with DualDn (RAW) and with our method with 25/15/9 neighbors.

that DualDn is a modular RAW-to-RGB scheme that denoises both before and after demosaicking, and its reported metrics are those of the first RAW-to-RAW module, which uses Restormer (Zamir et al., 2022) as a backbone. We also include quantitative comparisons of our single-scale variant with 15 neighbors against lighter baselines: BM3D, DnCNN and N3Net. Each of the 50 test images of the dataset are assigned 20 bounding boxes that delimit the regions that are used for benchmarking; we denoise each of these crops individually, using the noise parameters embedded in the full image metadata to generate their noise maps.

Table 10 reports the average PSNR and SSIM in the sRGB domain after applying the benchmark's processing pipeline on the denoised images. Our lightweight single-scale variant already outperforms BM3D,

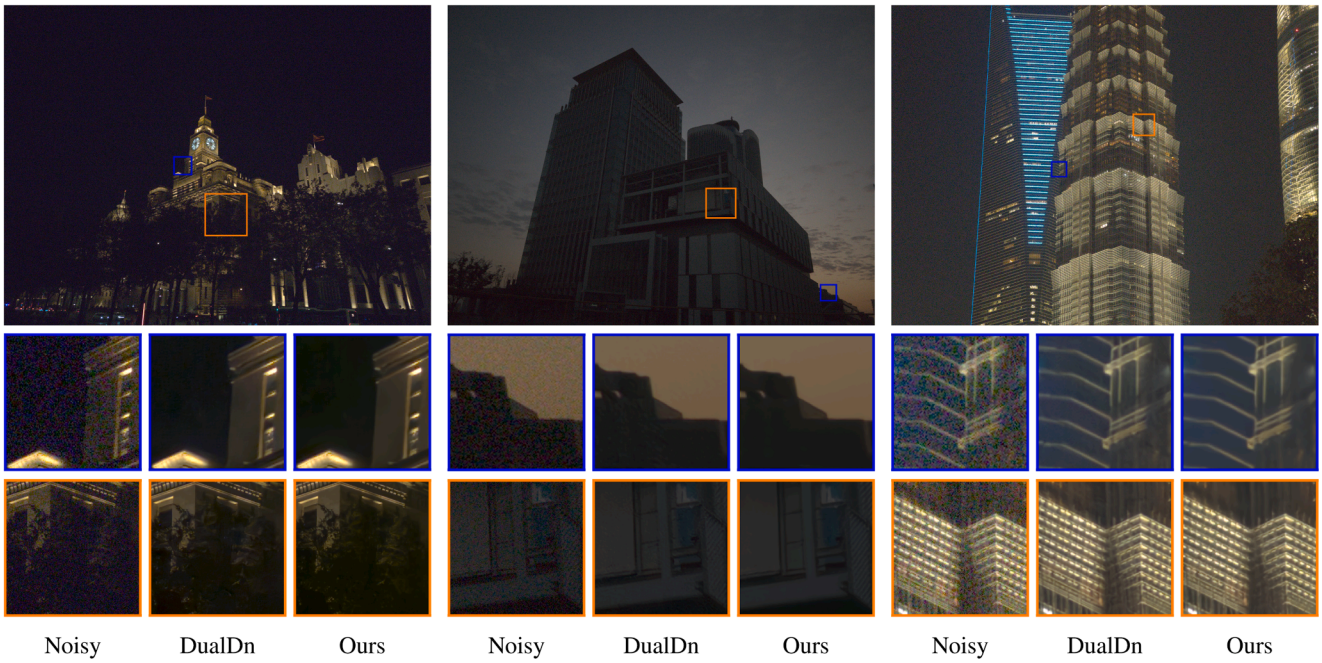


**Fig. 13.** Visual comparison between DualDn (RAW) and our method with 25/15/9 neighbors on low-light outdoor photographs. DualDn leaves residual grain and low-frequency noise on dark flat regions, and shows a slight color cast around dark textures, which we successfully correct, although at the cost of slightly more smoothed results.

DnCNN and N3Net by over 1.6 dB of PSNR, demonstrating that the proposed nonlocal block can be suitable by itself for low-resource environments. Our complete multiscale network surpasses UPI, CycleISP and PseudoISP, showing a notable PSNR advantage. DualDn achieves the highest PSNR—consistent with Restormer's results in the AWGN setting—but shows some regression in terms of SSIM, where we attain a higher value. Visual inspection of the two crops in Fig. 11 (extracted in sRGB from the web platform) reveal some of DualDn's limitations that may partially explain the metrics' discrepancies. Residual noise persists in darker flat regions, being especially noticeable at the bottom right of the blue patch. This problem is also present in UPI, where the grain is even more discernible. Also in that same patch, a hot-pixel artifact remains visible above the leftmost white region in both UPI and DualDn. CycleISP does not leave residual noise and corrects the artifact, but has trouble reconstructing straight edges, bending them slightly. Our method completely eliminates noise and the artifact, and produces a visually pleasant reconstruction. Images from PseudoISP are unavailable. In the orange patch, it can also be seen that all denoisers (especially CycleISP) exhibit a slight color bias toward red in what should be a uniformly black sofa. This is low-frequency residual noise that is common in dark regions with low signal-to-noise ratio. Our method substantially attenuates this bias, yielding more visually consistent results.

### 5.2. Results on in-the-wild images

The images in DND have been captured in a controlled setting, where most of them display bright, daylight scenes and are corrupted with a relatively low noise level. As consequence, results in the benchmark may not be representative of a method's capability of denoising daily life photographs. To assess generalization to a bigger variety image conditions, we perform a visual evaluation on the in-the-wild dataset collected by Li et al. (2024b). This dataset comprises high-resolution noisy RAW images captured by smartphone cameras of three different brands with a diversity of ISO values, of which a ground truth is not available.

Although the image files contain metadata with noise information at the sensor, we find that the embedded curves are probably miscalibrated and strongly overestimate the level of noise in some sensor settings. As such, we use the same algorithm as in Section 3.3 to reestimate the curves for every image. The first row of Fig. 12 shows the curves for an image taken with each camera. The solid red line is constructed from the parameters in the image metadata, while the dotted black one is our estimation from the black points obtained using Colom and Buades (2013). It is apparent that the camera curves are uniformly above our estimations.

Below the plots in Fig. 12, we showcase the results of denoising the images with DualDn (26.5M parameters) and our method (15.3M parameters) using the estimated black dotted curves. As the commercial ISP on the devices is unknown, we apply a simple processing pipeline using the EXIF metadata for the purpose of visualization. The Huawei scene is notably dark and thus serves to highlight the behavior of both methods in low-light conditions. DualDn exhibits difficulties in such dark regions: blotchy artifacts appear around edges, blue-tinted low-frequency residual noise remains in flatter areas, and blue hot-pixel defects are visible. In the blue patch, the parasol loses its silhouette and blends into the dark background; in the orange patch, a blotch appears along the edge of the archway. Our method produces a cleaner reconstruction in both cases. On the Xiaomi image, we achieve finer texture preservation overall: DualDn oversmooths the wood markings and introduces yellowish artifacts around the newspaper text, whereas we recover sharper forms. Finally, on the iPhone photograph, DualDn leaves noticeable residual noise on flat regions. While this underfiltering occasionally preserves local contrast (e.g. the edges of the plant appear slightly sharper), the persistent noise degrades the overall visual quality compared to our cleaner restoration.

Fig. 13 displays more examples of very low-light photographs from the same dataset, taken in outdoor settings with high ISO. Our method manages to successfully remove the noise everywhere with minimal texture loss, yielding visually faithful results even in regions with very low signal-to-noise ratio. This is noticeable in the orange patch of the left image, where noise obfuscates most of the trees. Where DualDn enshrouds them with a slight color bias in the near-black, we get a sharper reconstruction of the branches with a more cohesive color. Differences between how both methods treat dark regions can also be seen in the

blue patch of the same image: DualDn leaves some mix of grain and low-frequency noise in the sky and in the wall of the building, which we completely filter. This residual noise left by DualDn is mostly found in dark, flat regions, as further showcased in both patches of the middle image, and follows the patterns already seen in the third image of Fig. 12 in an indoor setting. Our method again overcomes this issue, fully removing the noise in such regions. However, we find a limitation in how the network oversmoothes subtle textures inside these dark, flat areas. As an example, in the right orange patch, the details in the shadowed corner intersecting the two walls of the building, as well as the dimmer levels at the top, are better preserved by DualDn. On the contrary, the brighter grid-like lights are appropriately reconstructed by both methods. The same behavior is observed in the right blue patch, again presenting a heavier smoothing on the gray texture to the right, although in this case DualDn attains a rougher reconstruction with some residual noise.

Overall, these experiments confirm that the proposed method generalizes well across sensors, lighting conditions and ISO levels, yielding visually consistent, sharp results without residual noise or excessive texture smoothing, and being competitive with larger, more complex RAW denoising pipelines in terms of image quality—traits that make it suitable for mobile photography.

## 6. Conclusion

We have presented a neural network architecture that translates the classical three-step pipeline of nonlocal patch-based image denoising methods (matching, collaborative filtering, and aggregation) into a fully learnable framework operating on feature representations. The centerpiece of the network is a Nonlocal Feature Matching and Filtering Block, which uses a convolutional neural network to select a fixed number of neighbors as support for subsequent filtering through the modulation of a groupwise linearly transformed feature stack. This block is placed within a UNet, efficiently expanding the receptive field in a nonlocal manner without requiring expensive self-attention or excessive depth.

By means of training data selection and using a noise level map as input, the proposed network is designed for the task of RAW-to-RAW image denoising. We have curated a dataset of clean RAW images alongside a list of noise profiles for a variety of camera sensors and ranges of ISO values, which we use to synthesize realistic RAW images corrupted with Poisson-Gaussian noise. Training the network on this data yields a sensor-agnostic denoiser that generalizes well to unseen devices and lighting conditions, as evidenced by visual results on in-the-wild photographs. Quantitative experiments also show that the proposed method achieves results competitive with state-of-the-art CNN and transformer-based denoisers while using significantly fewer parameters.

While this work focuses on single-image denoising, the generic structure of the nonlocal block naturally lends itself to video applications. Extending the neighbor search to nearby frames would allow the network to exploit temporal redundancy, a promising direction for future work.

## CRediT authorship contribution statement

**Marco Sánchez-Beeckman:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation; **Antoni Buades:** Writing – review & editing, Supervision, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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